

Business Analysis Associate

Accredited by



Bi-variate Analysis

Types of Bivariate Analysis

- Continuous-Continuous: Analyzing the relationship between two continuous variables, typically done using correlation or regression analysis.
- Categorical-Categorical: Analyzing the association between two categorical variables using techniques like contingency tables or chi-squared tests.
- Continuous-Categorical: Examining the impact of a continuous variable on a categorical variable using methods such as t-tests or ANOVA.

Bivariate Analysis: Continuous & Continuous

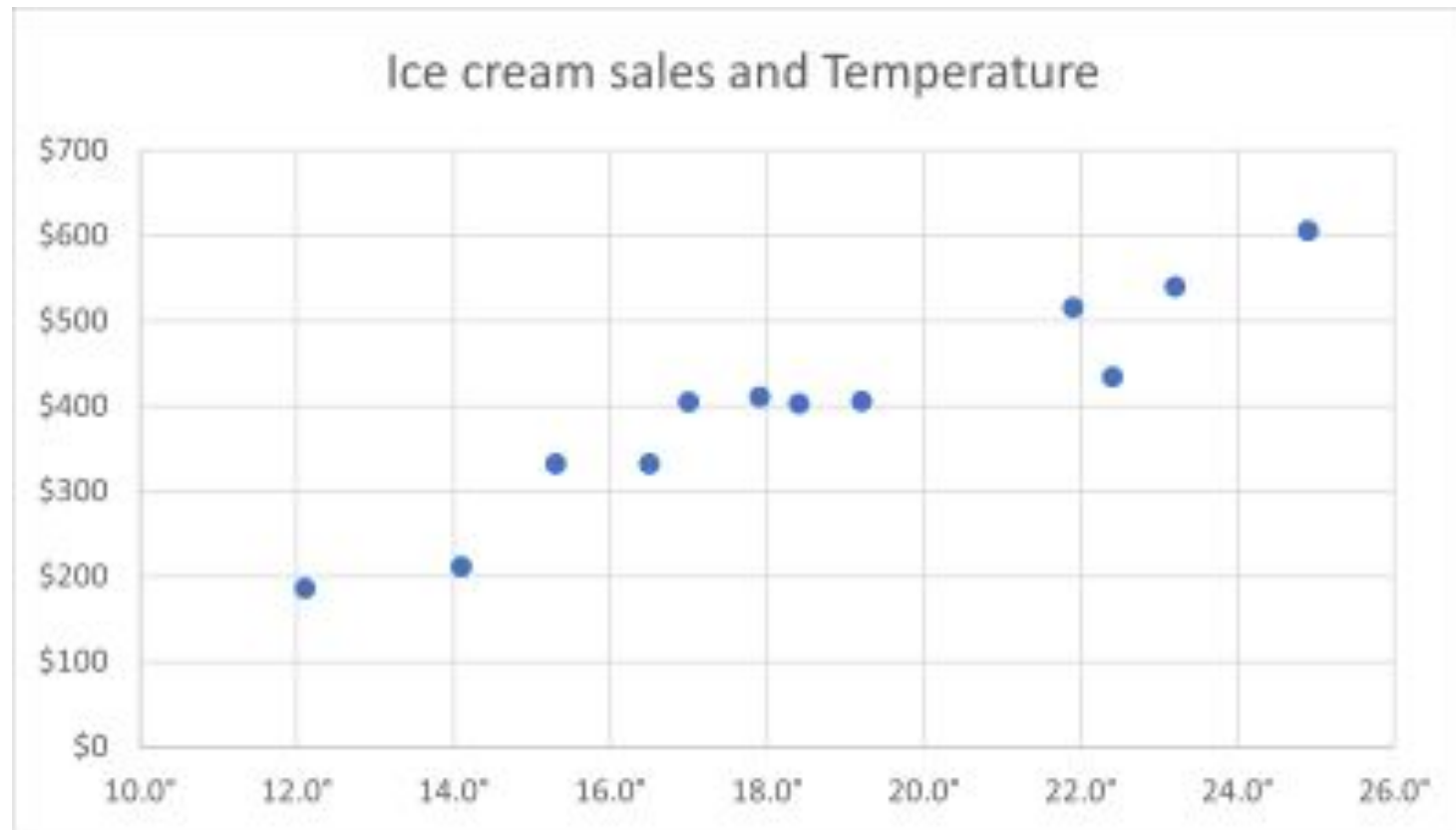
Some of the common types of bivariate analysis include:

- Scatter Plots
- Regression Analysis
- Correlation Coefficients



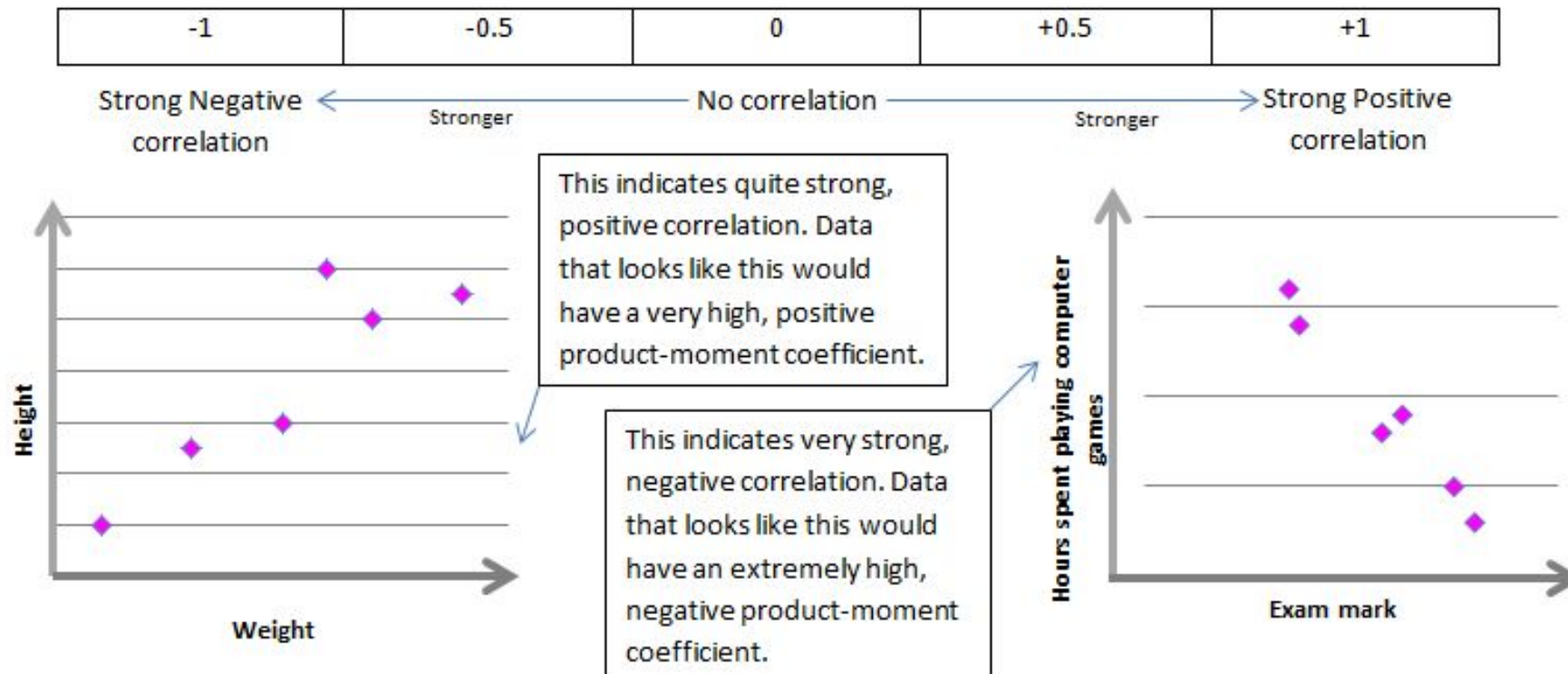
Scatter Plots

Scatterplot provides you with a visual idea of the pattern that your variables follow.



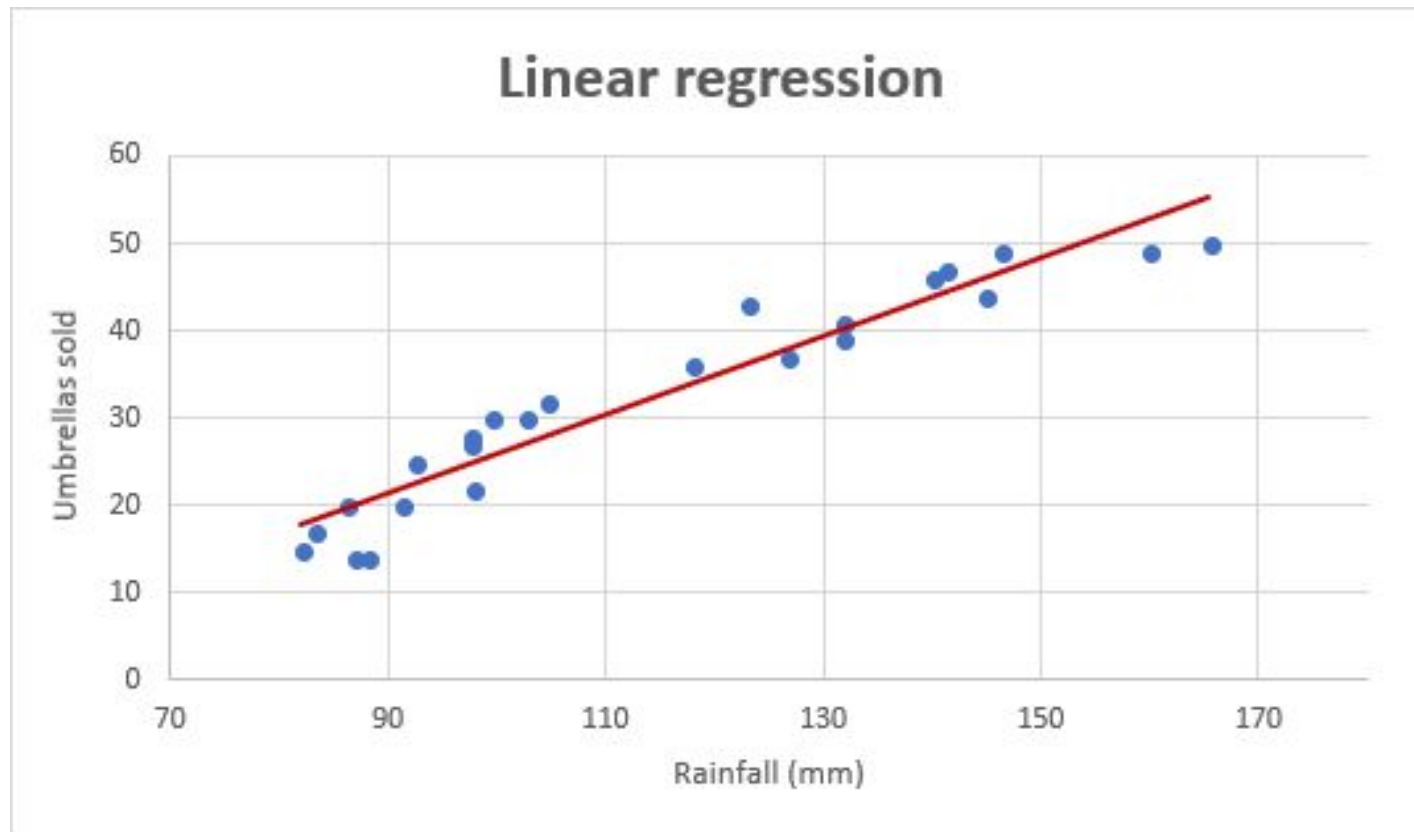
Correlation Coefficients

Calculation of values for correlation coefficients are performed using a computer, although here, you can find the steps to find the correlation coefficient by hand. This coefficient acknowledges you if the variables are related.



Regression Analysis

Regression analysis is a catch-all term for a wide variety of tools that can be used to determine how your data points might be related. The points in the image above seem like they could follow an exponential curve (as opposed to a straight line).



Time Series Data

- Time series data is a sequence of data points that measure some variable over ordered period of time. It helps to understand and forecast data patterns. It is important to look for the pattern or components in a Time Series data such as Trend, Seasonality etc.
- Examples: Weather Forecasting, Sales Trends, Data captured in CCTV, Stock price, etc

Trend Analysis

- **Trend** is a pattern in data that shows the movement of a series to relatively higher or lower values over a long period of time. In other words, a trend is observed when there is an increasing or decreasing slope in the time series.
- Hands-on – Trend Analysis for a Time Series Data.

Line Chart:

- Show how a continuous variable changes over time.

Types of Trend

- **Upward trend:**

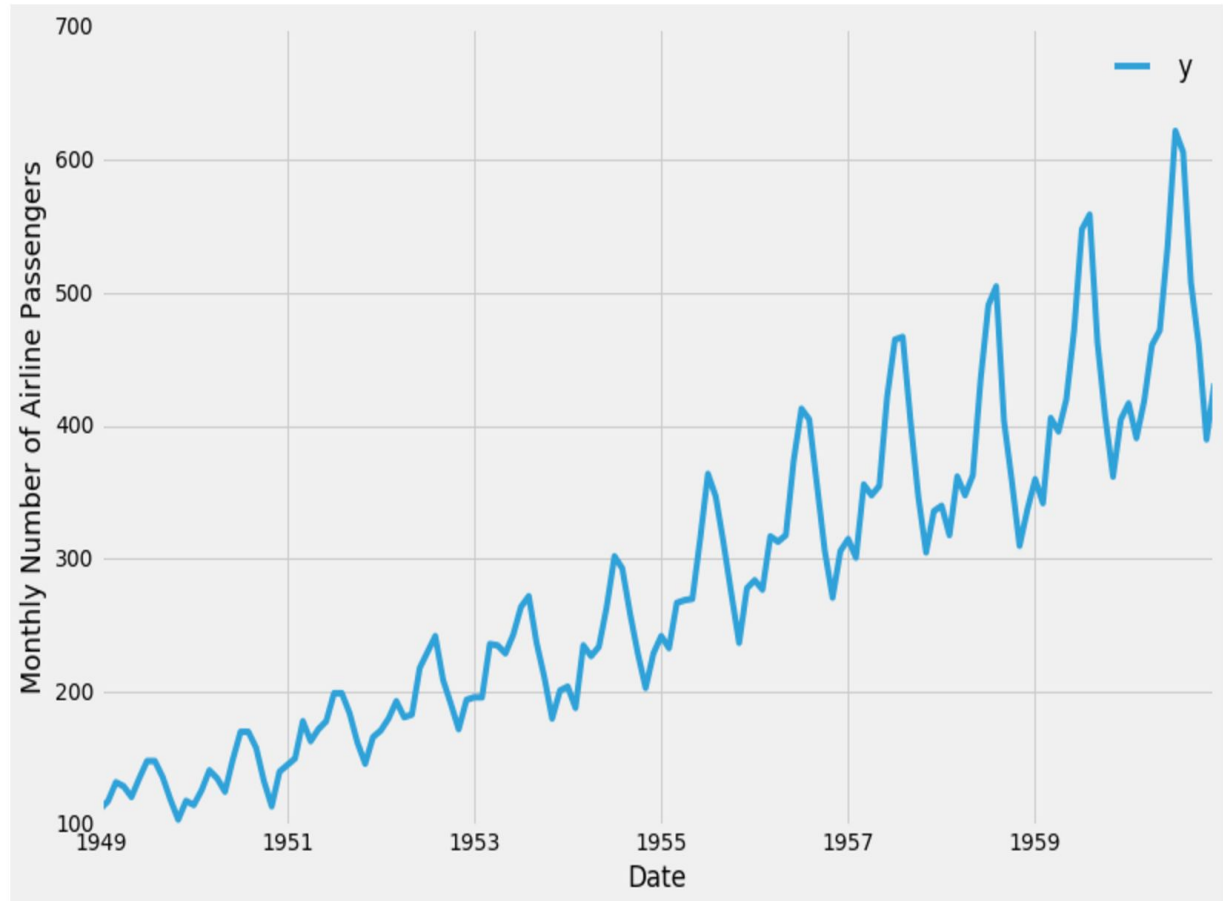
Time Series Analysis shows a general pattern that is upward then it is Upward trend.

- **Downward trend:**

Time Series Analysis shows a pattern that is downward then it is Downward trend.

- **Horizontal or Stationary trend:**

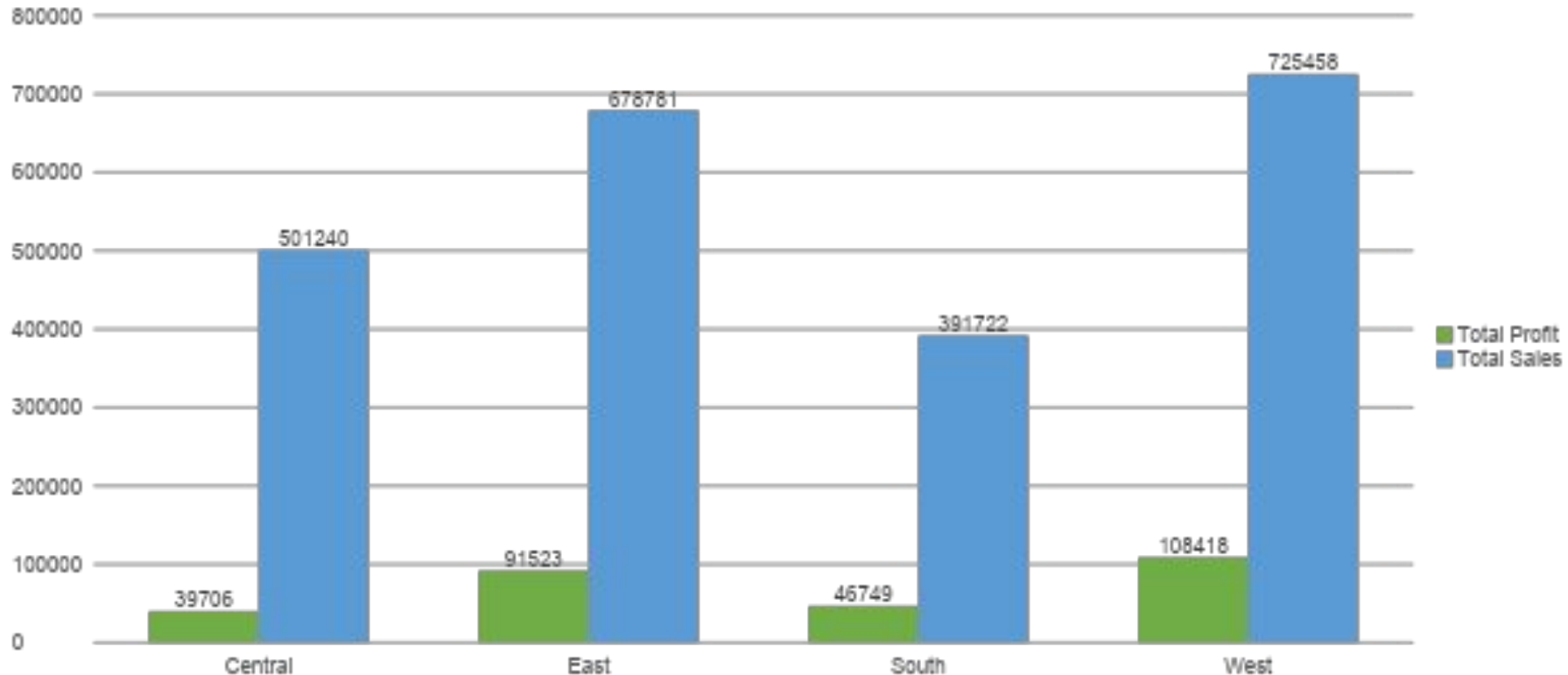
If no pattern is observed then it is called a Horizontal or stationary trend.



Comparison Analysis

Comparison analysis involves comparing different variables, groups, or scenarios to identify similarities, differences, patterns, and trends.

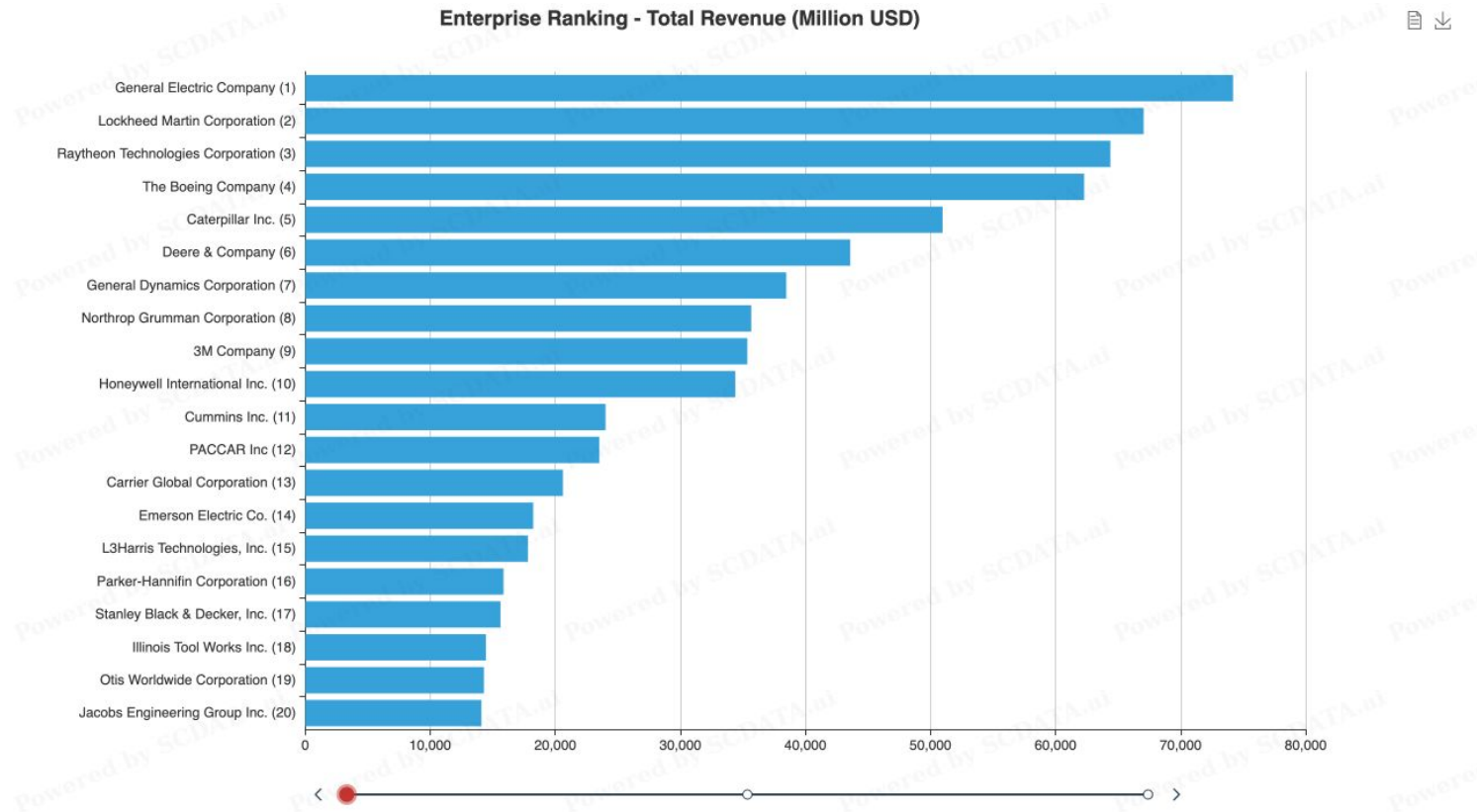
Eg. Compare sales and profit across all the regions.



Ranking Analysis

Process of evaluating and analyzing the rankings of items, entities or data, based on certain criteria or attributes.

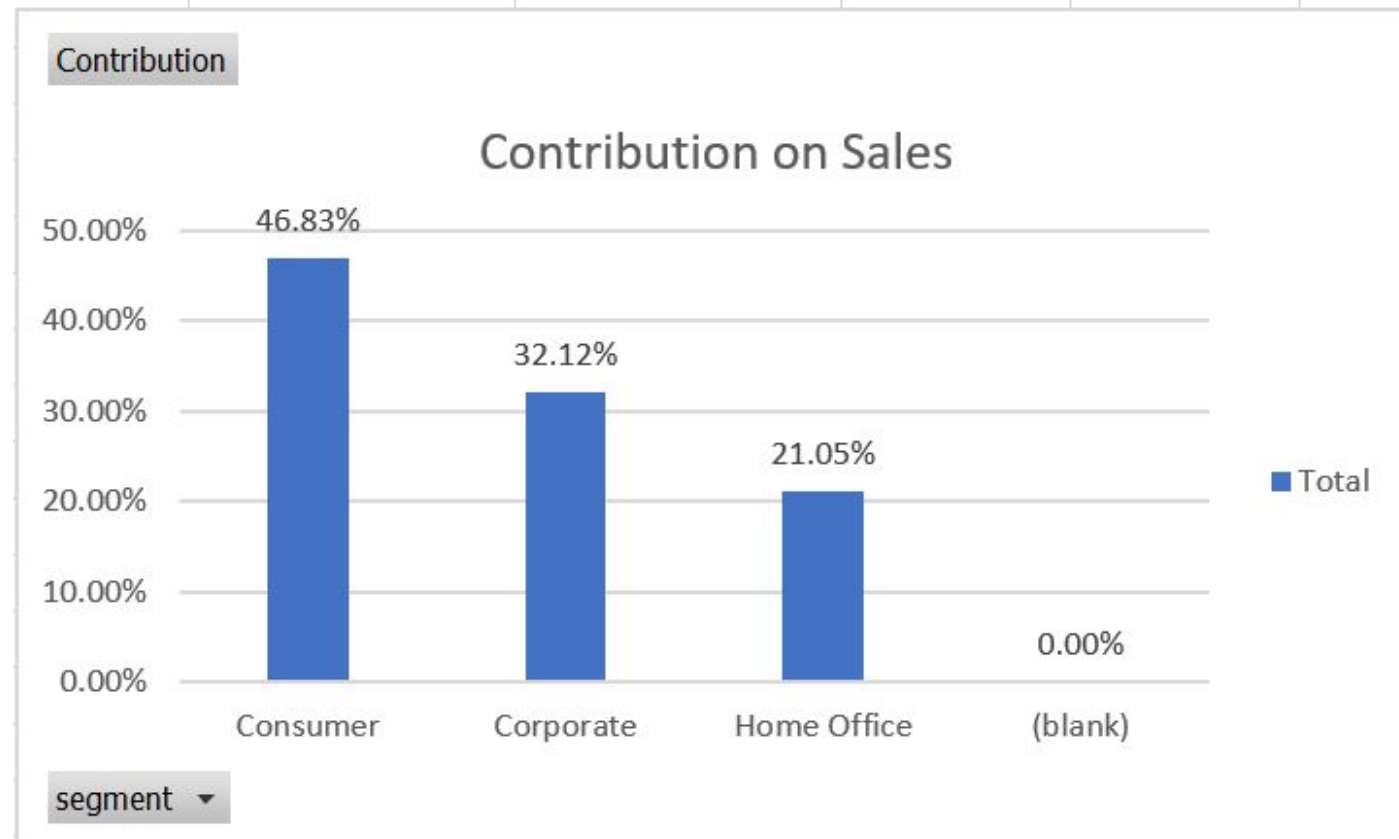
Eg. Find the top 5 products in terms of Sales.



Contribution Analysis

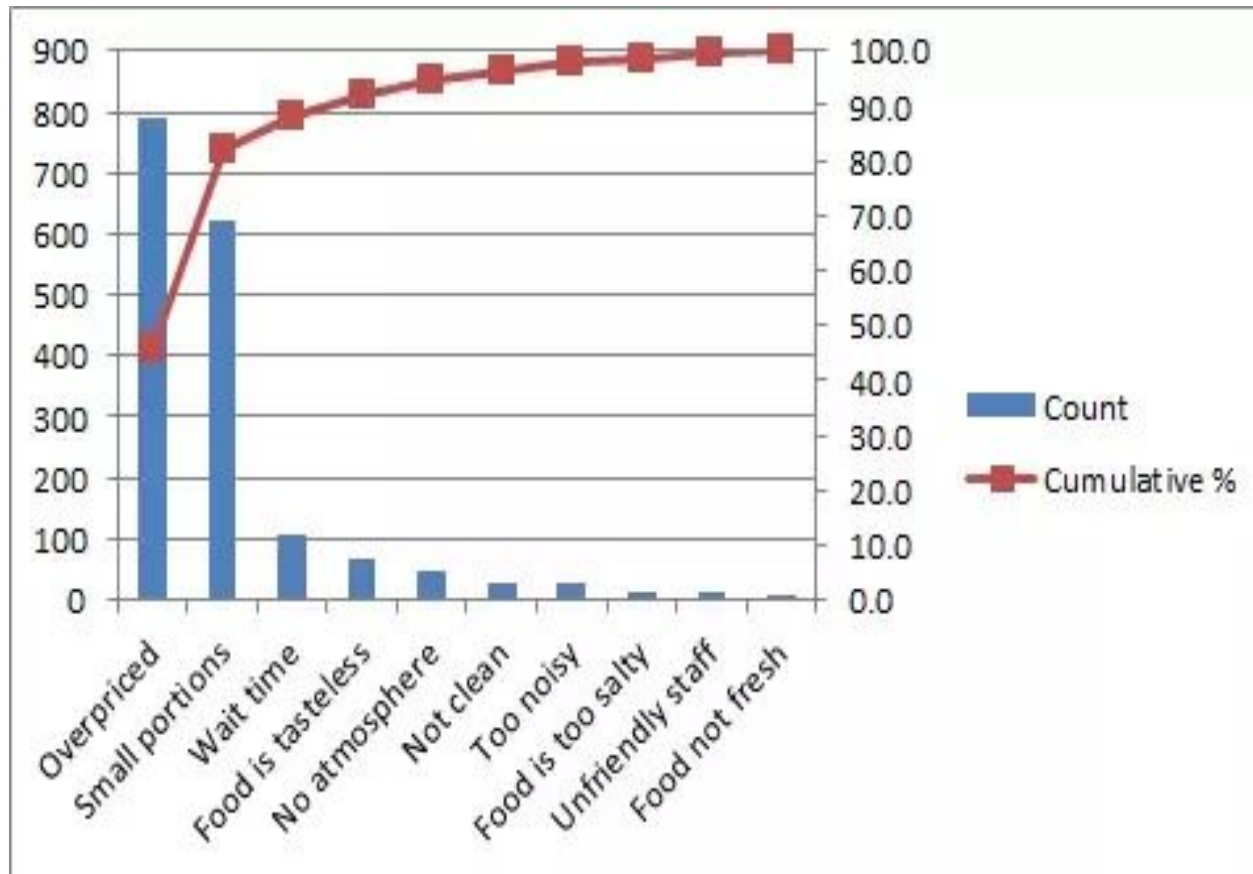
Contribution analysis is a powerful tool that helps businesses and organizations to analyze the contribution of various factors towards a particular outcome.

Eg. Analyze the contribution of categories towards Sales, Revenue, etc



Pareto Analysis

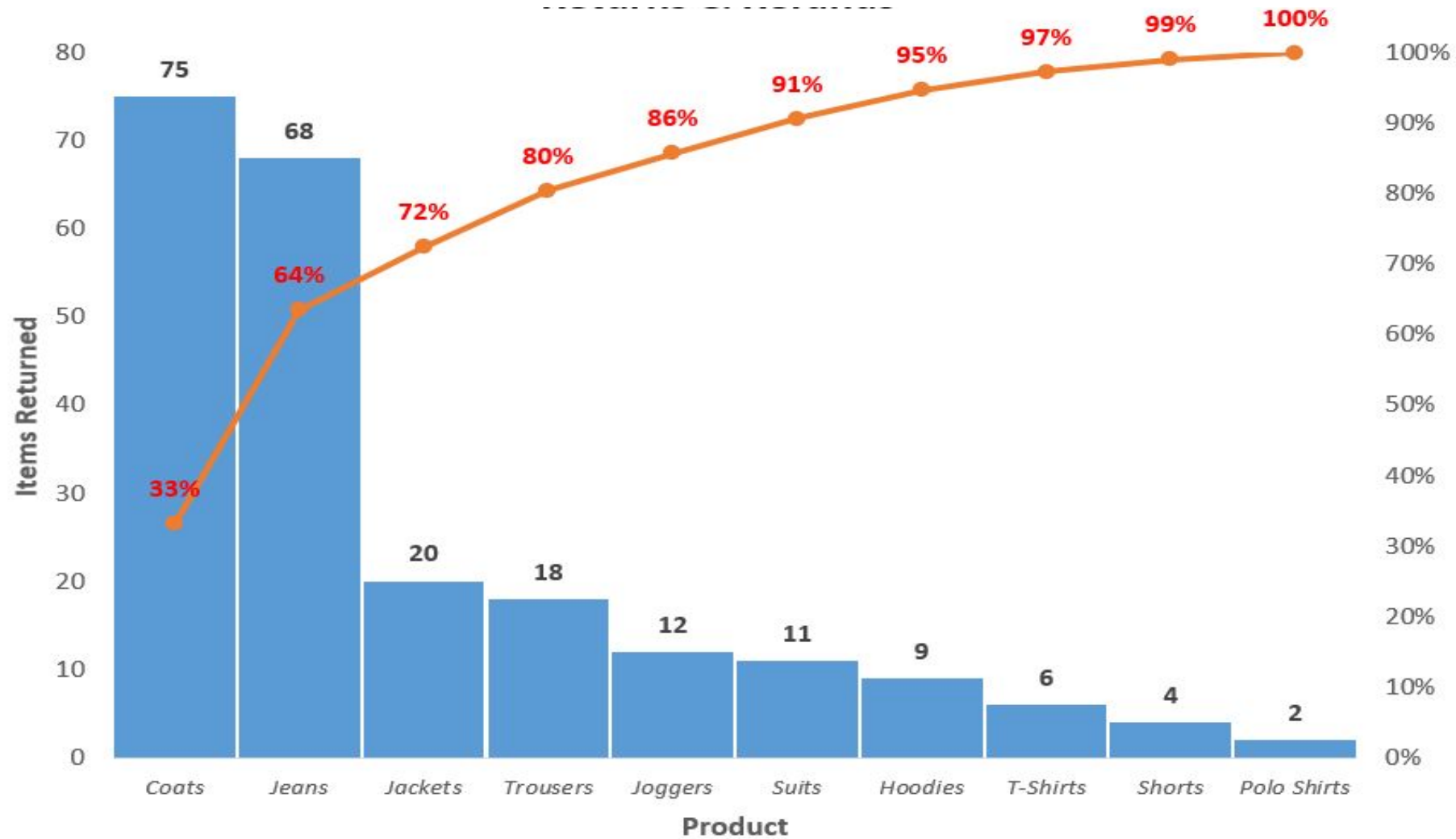
Pareto analysis, also known as the 80/20 rule, is a management tool used to identify the most significant factors or causes that contribute to a particular outcome. In Pareto analysis, it is assumed that 80% of the effects are caused by 20% of the factors.



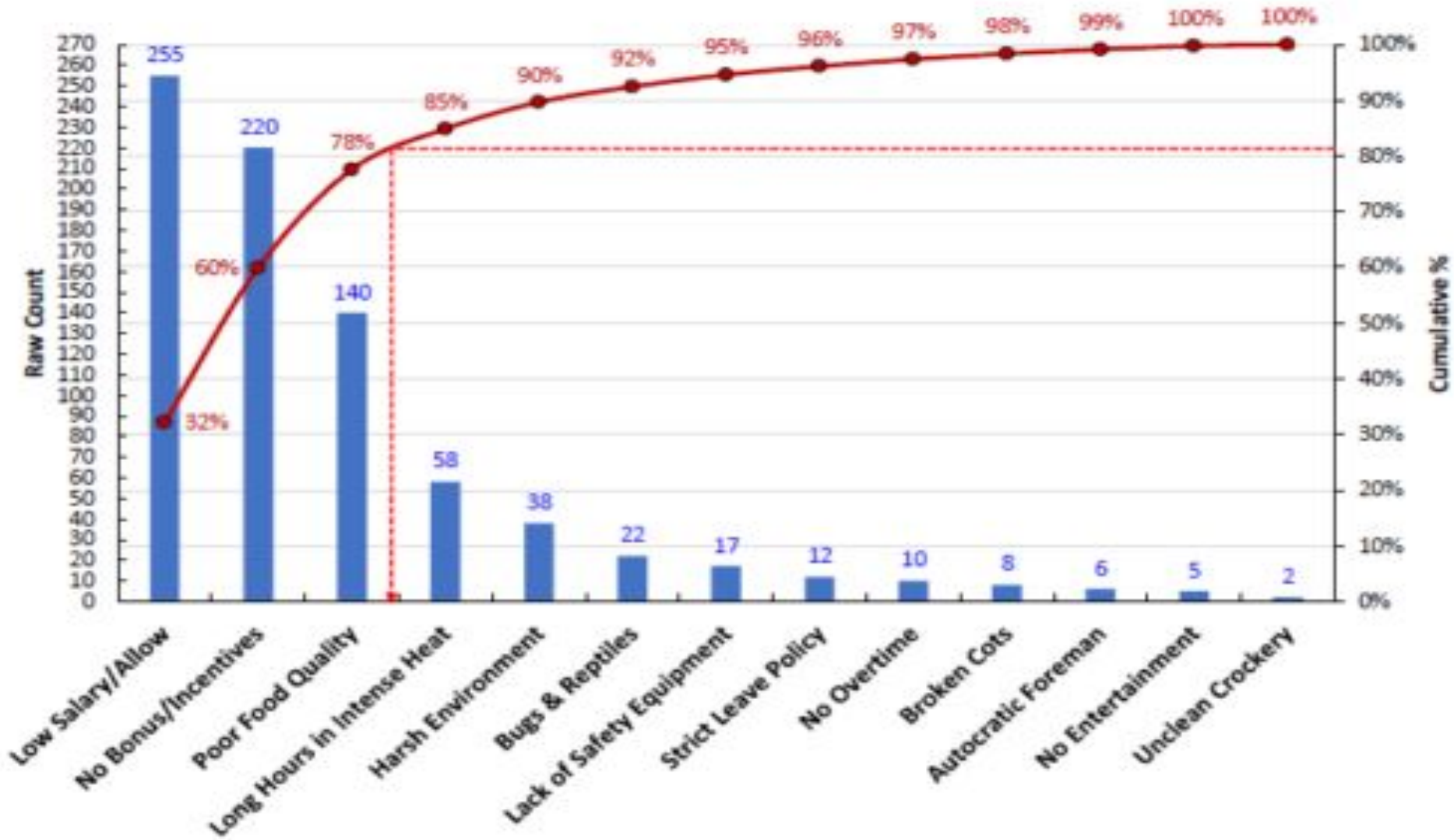
Restaurant

- Also called an 80-20 rule, 80% of the problems are caused by 20% of facts whether if you look into profits, customer complaints, technical issues, product defects, etc.
- It statistically separates a limited number of input factors that have a greater impact on an outcome
- It is a powerful quality and decision-making tool.
- It organizes and displays information to show the relative importance of various problems or causes of problems.

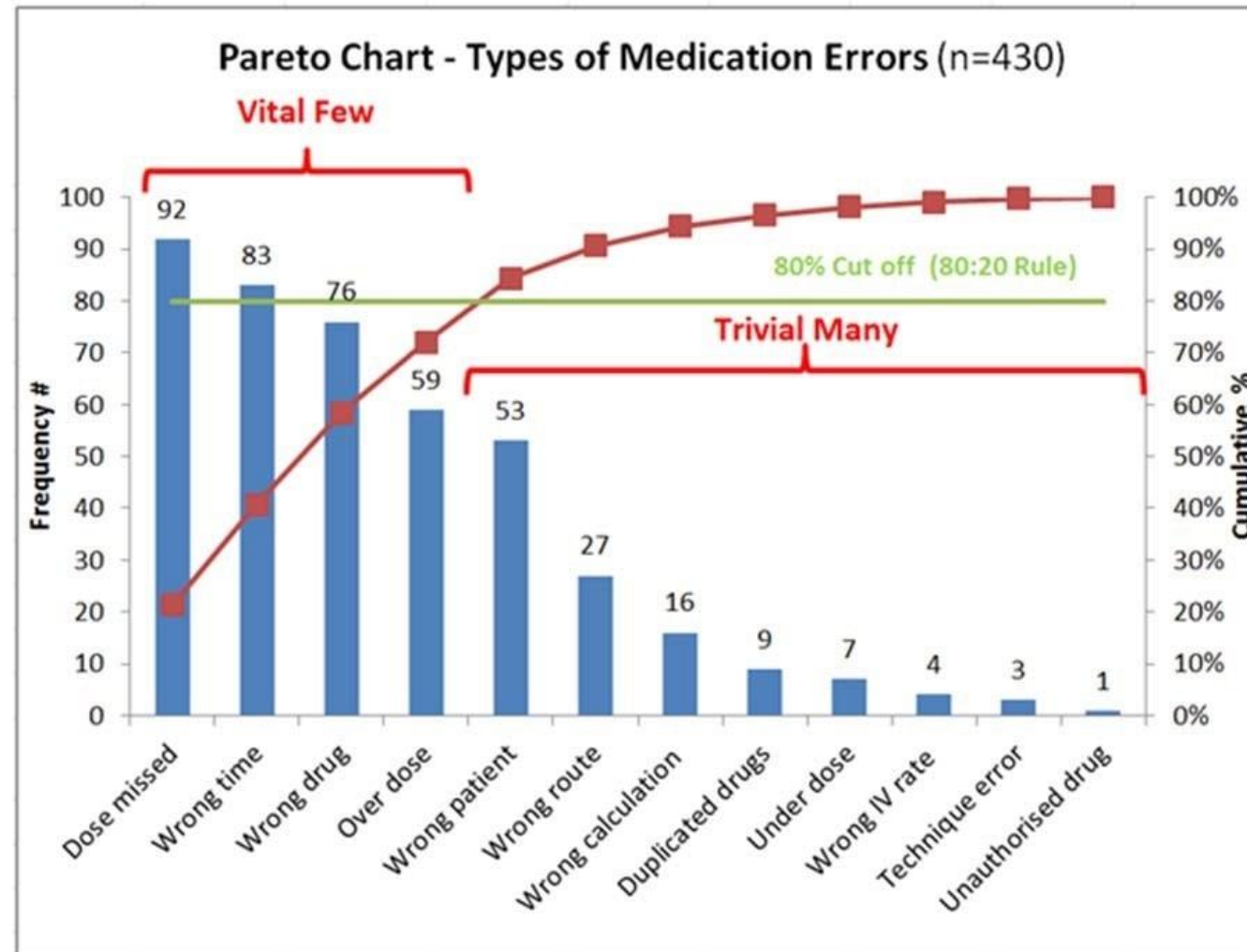
Returns & Refunds



Attrition



Medication Errors



Interactive Dashboards



Interactive dashboards in Excel are a powerful way to present and analyze data. They allow users to explore and analyze data through interactive charts, tables, and visualizations.

It is a Data Management tool which comes with various interactive features where users can explore the data on a deeper level.

- Prepare your data.
- Create your dashboard with charts, tables or visualizations.
- Add interactivity using slicer, filters, pivot table etc
- Customize your dashboard

With interactivity, you can create a powerful and effective dashboard that will help you make better decisions based on your data.

Refer & Earn

up to ₹ 5,000/- per referral



End of the Module

 **Datamites**

Contact care@datamites.com for any queries

